SEOULTECH

Lecture 12
Mobile Networks I

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CIEL Communication & Intelligence Laboratory
통신및지능연구실

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Course Schedule (Tentative)

- **F2F**: Face-to-face, **Recorded**: Recorded Lecture Video

W	Topics	Note	Date
1	Orientation and Basics of data communications and networks	F2F Lecture & Recorded	09/03
2	Network structures & models	Recorded & F2F Lecture	09/10
3	Introduction to physical layer I: data and signals	Recorded & F2F Lecture	09/17
4	Introduction to physical layer II: transmission & performances	Recorded & F2F Lecture	09/24
5	Digital transmission & Analog-to-digital conversion	Recorded & F2F Lecture	10/01
6	Analog transmission & Multiplexing	**No F2F Class (Holiday), Only Recorded**	**10/08**
7	Transmission Media, Switching & Performances	Recorded & F2F Lecture	10/15
8	**Midterm exam**	**Class hours**	**10/22**
9	Basics on data-link & upper layers	Recorded & F2F Lecture	10/29
10	Error detection and correction	Recorded & F2F Lecture	11/05
11	Data link control & medium access control	**No F2F Class, Only Recorded**	**11/12**
12	Wired LAN (Ethernet), Wireless LAN (WiFi)	Recorded & F2F Lecture	11/19
13	Mobile networks I	Recorded & F2F Lecture	11/26
14	Mobile networks II & Other wireless networks	Recorded & F2F Lecture	12/03
15	**Final exam**	**Class hours**	**12/10**

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OUTLINES



- Basics on mobile networks
- From 1G to 5G
- Fundamentals for mobile networks
- Summary & Next class

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Basics on Mobile Networks

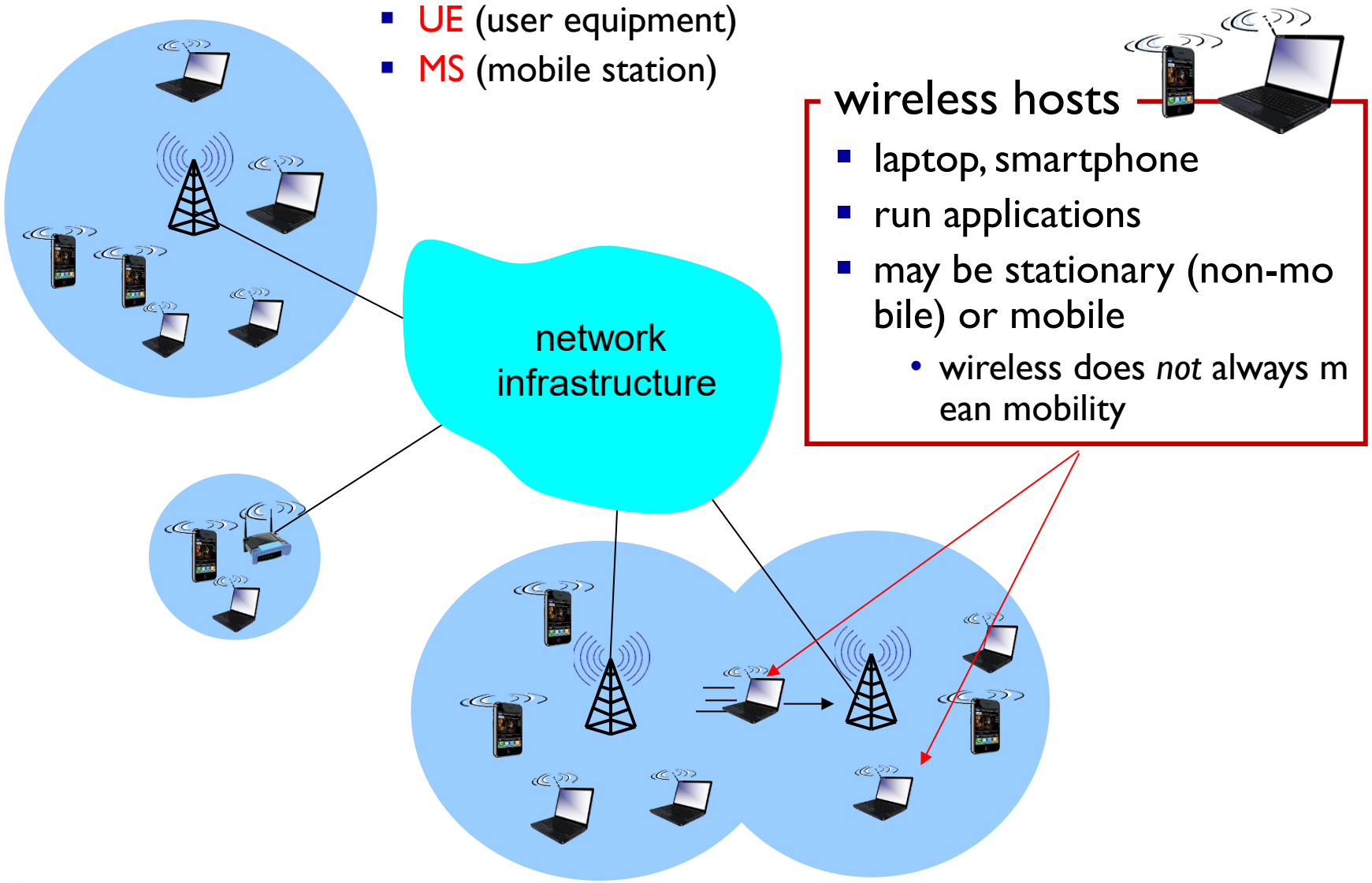
: Wireless network elements, link, architecture

- Basics on mobile networks
- From 1G to 5G
- Fundamentals for mobile networks
- Summary & Next class

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Elements of A Wireless Network

- UE (user equipment)
- MS (mobile station)



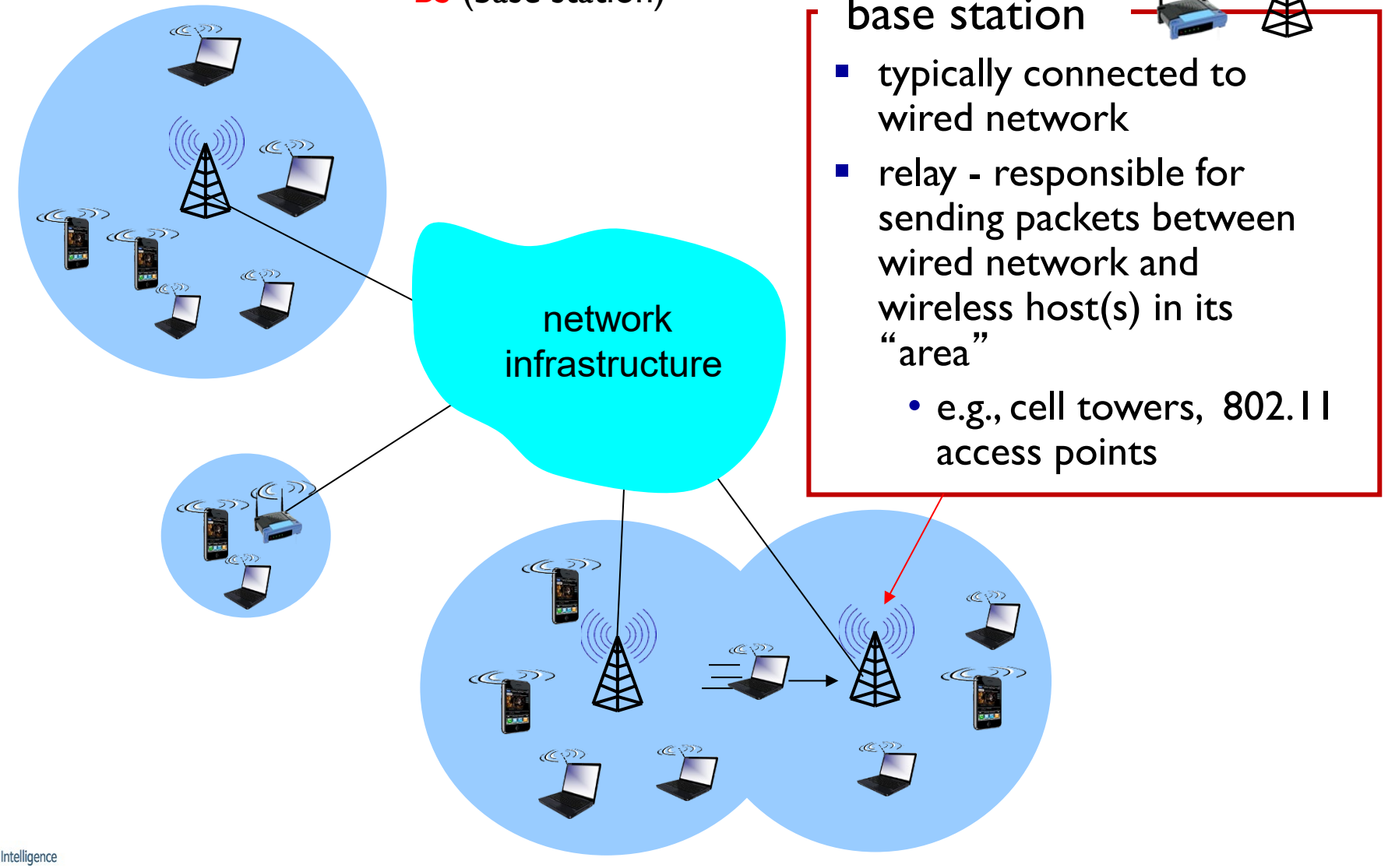
wireless hosts

- laptop, smartphone
- run applications
- may be stationary (non-mo bile) or mobile
  - wireless does *not* always m ean mobility

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Elements of A Wireless Network

- BS (base station)

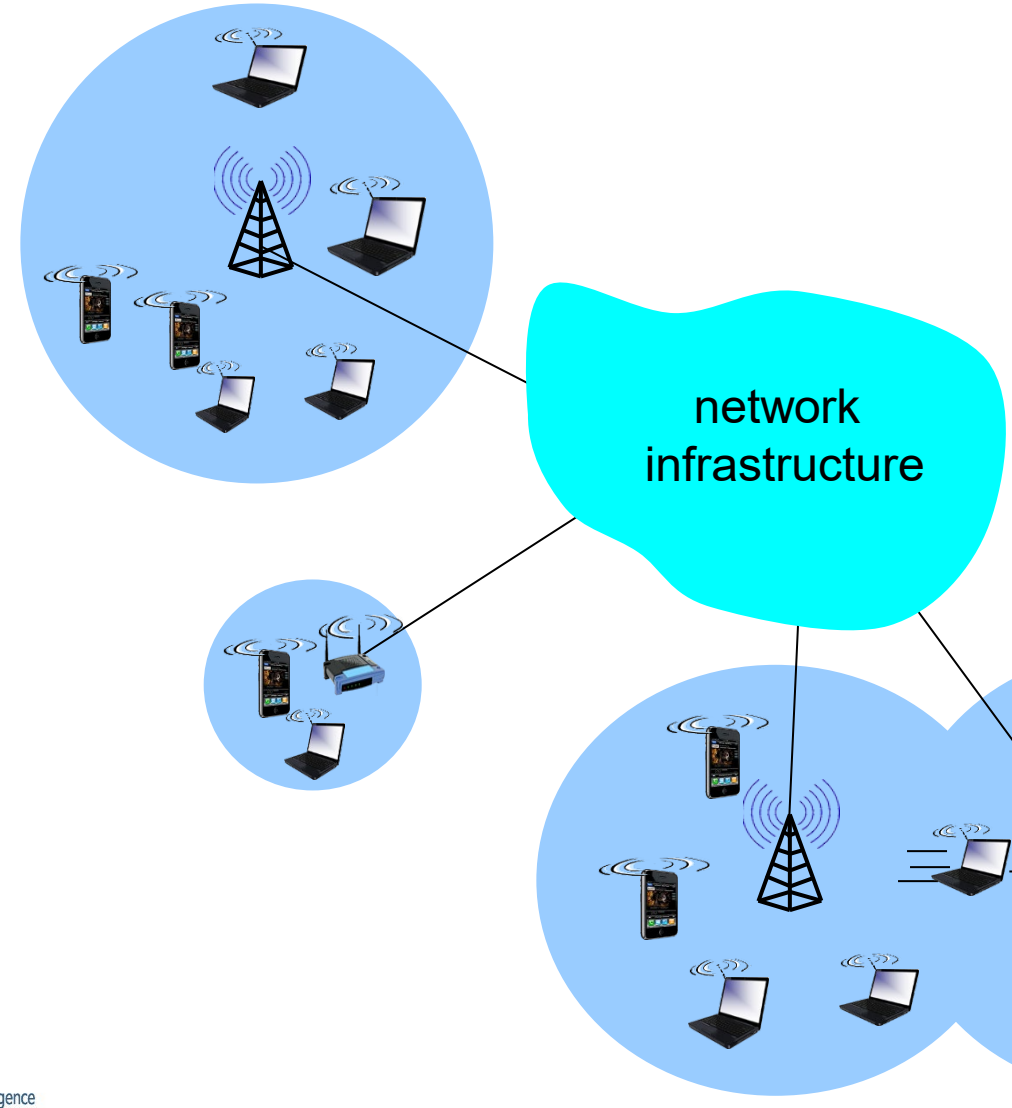


base station

- typically connected to wired network
- relay - responsible for sending packets between wired network and wireless host(s) in its "area"
  - e.g., cell towers, 802.11 access points

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Elements of A Wireless Network



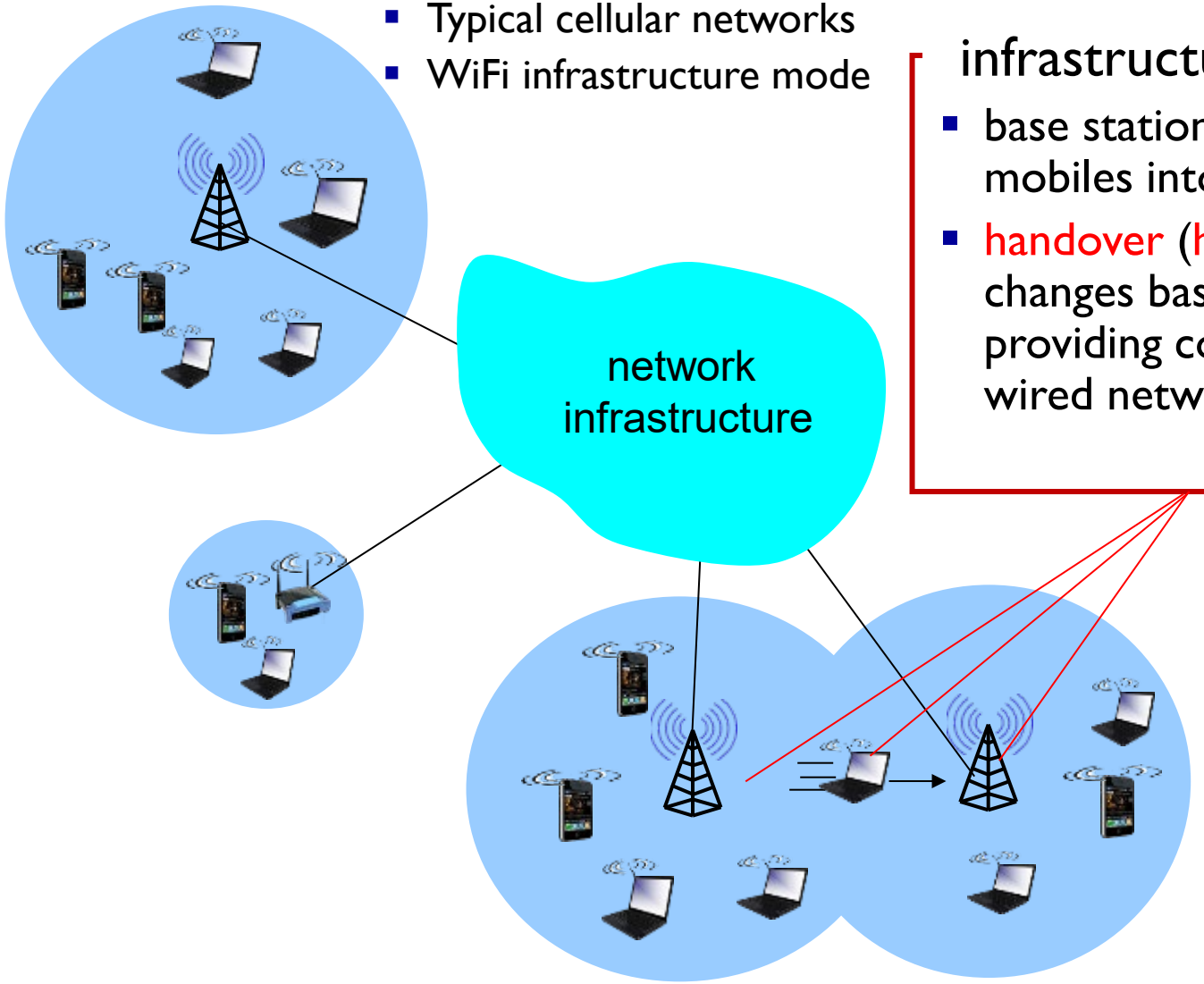
wireless link

- typically used to connect mobile(s) to base station
- also used as backbone link
- multiple access protocol coordinates link access
- various data rates, transmission distance

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Elements of A Wireless Network

- Typical cellular networks
- WiFi infrastructure mode



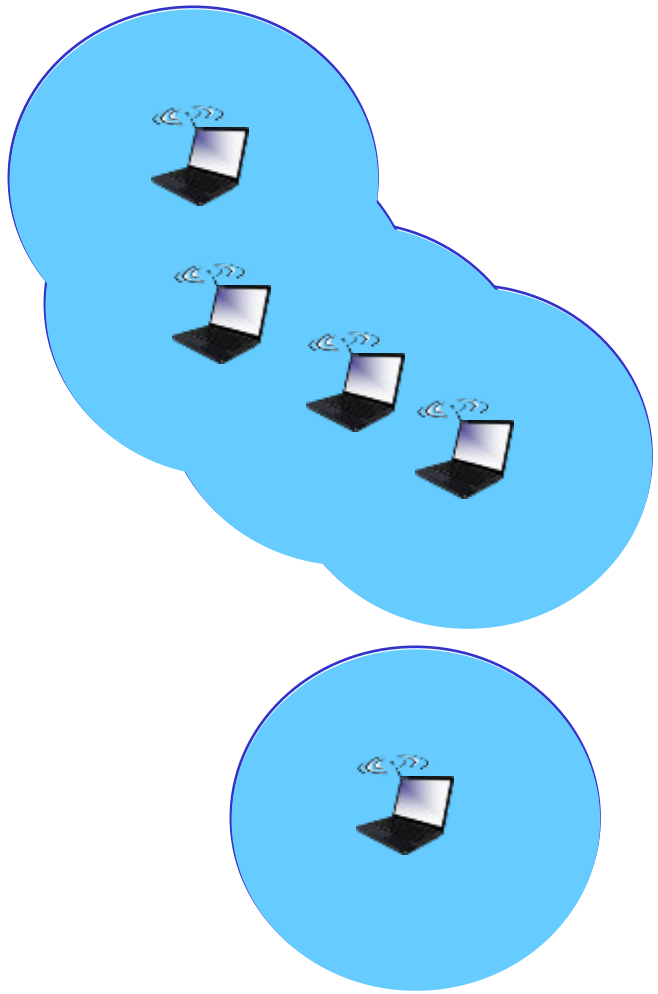
infrastructure mode

- base station connects mobiles into wired network
- handover (handoff): mobile changes base station providing connection into wired network

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Elements of A Wireless Network

- D2D (device to device) communications in cellular networks
- WiFi ad hoc mode



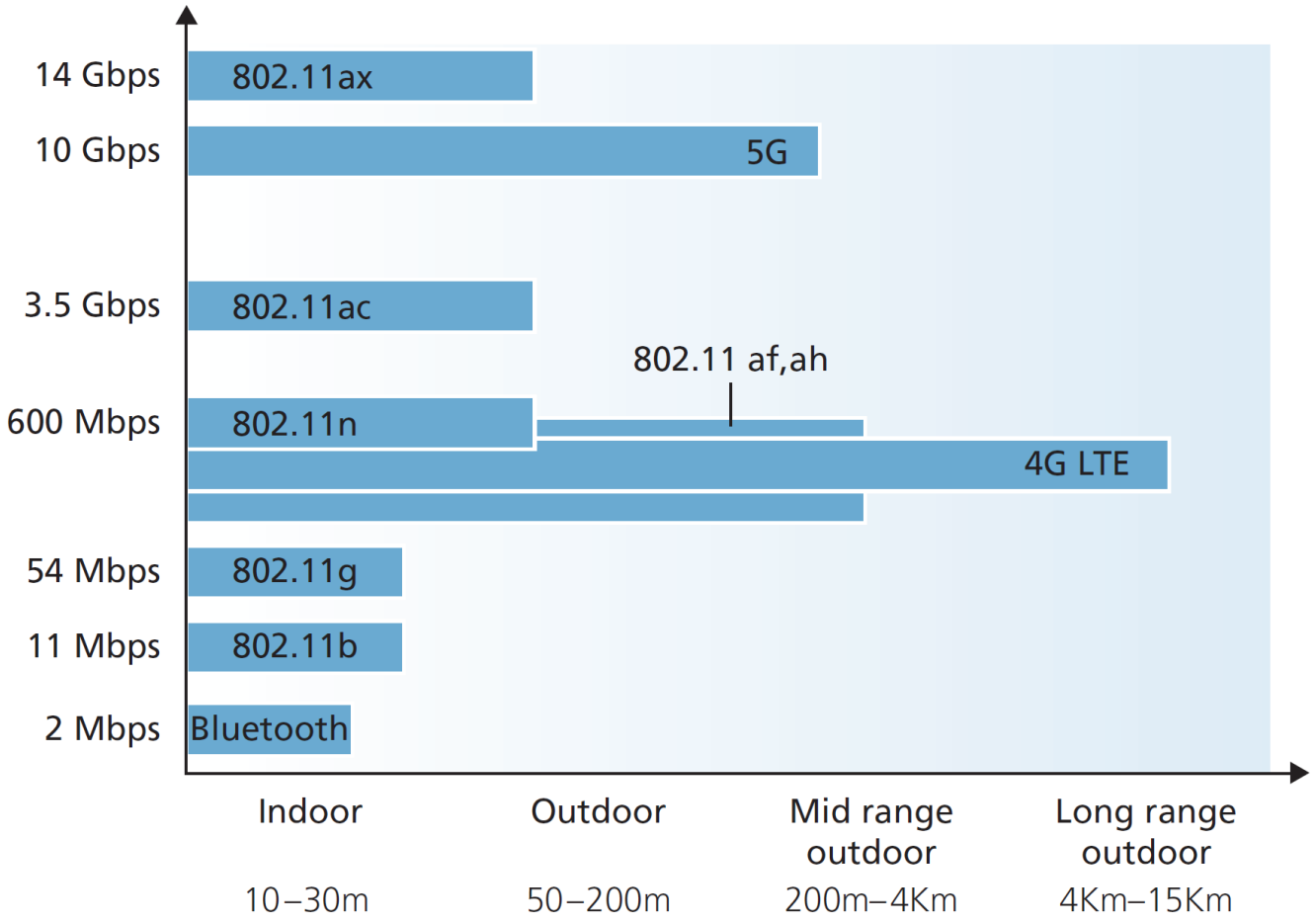
ad hoc mode

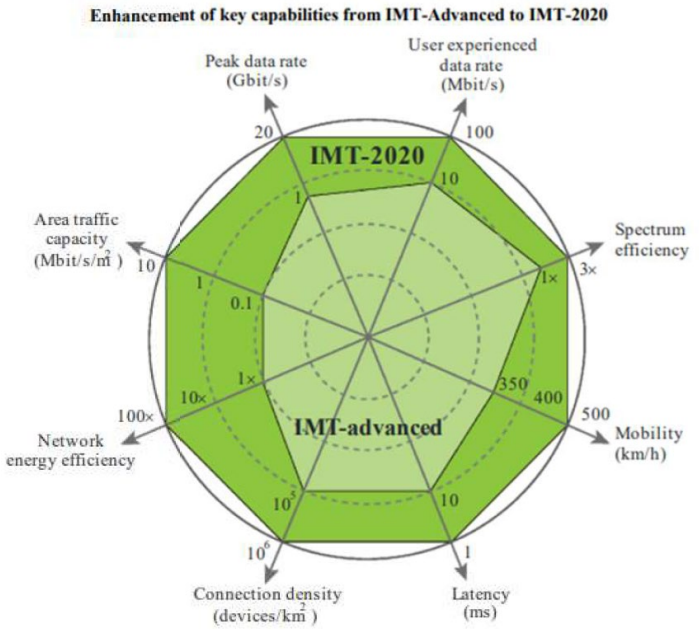
- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

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[Review] Wireless Transmission Rates and Range

- Wireless transmission rates and range
  - For WiFi, cellular 4G/5G and Bluetooth standards (note: axes are not linear)





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Wireless Network Taxonomy

	Single hop	Multiple hops
Infrastructure (e.g., BS, AP)	Host connects to base station (WiFi, cellular[4G,5G]) which connects to larger Internet	Host may have to relay through several wireless nodes to connect to larger Internet: *mesh net*
Infrastructure -less	No base station, no connection to larger Internet (Bluetooth, ad hoc nets)	No base station, no connection to larger Internet. May have to relay to reach other a given wireless node MANET, VANET

- MANET: mobile ad-hoc network
- VANET: vehicular ad-hoc network

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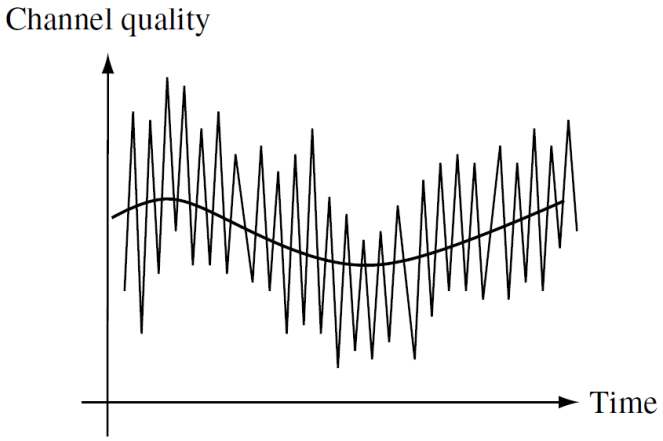
Wireless Link Characteristics

- Important differences from wired link
  - Decreased signal strength: radio signal attenuates as it propagates through matter (path loss)
  - Interference from other sources: standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
  - Multipath propagation: radio signal reflects off objects ground, arriving ad destination at slightly different times

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[Review] Wireless Channels

- Variations of channel strength over time and over frequency



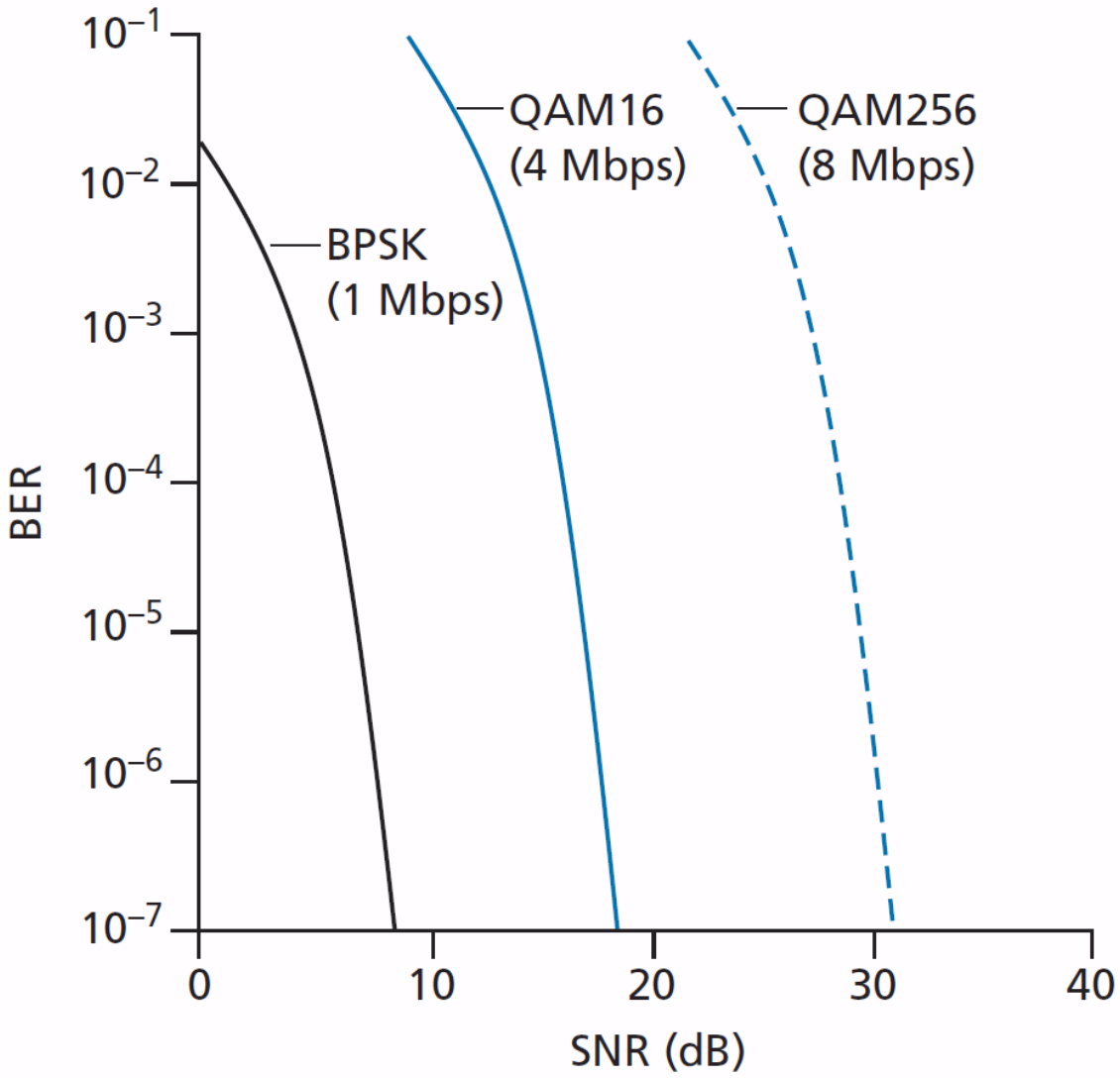
- Large-scale fading
  - Due to path loss of signal as a function of distance
  - Due to shadowing by large objects such as buildings and hills
  - Order of the cell size, and typically frequency independent

- Small-scale fading
  - Due to constructive and destructive interference of the multiple signal paths between transmitter and receiver
  - Order of carrier wavelength, and frequency dependent

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Wireless Link Characteristics

- SNR: signal-to-noise ratio
  - Larger SNR – easier to extract signal from noise (a "good thing")
- SNR versus BER tradeoffs
  - BER: bit error rate, PER: packet error rate
  - Given physical layer: increase power → increase SNR → decrease BER
- Given SNR: choose physical layer that meets BER requirement, giving highest throughput
- SNR may change with mobility: dynamically adapt physical layer (modulation technique, rate)
  - MCS: modulation and coding scheme



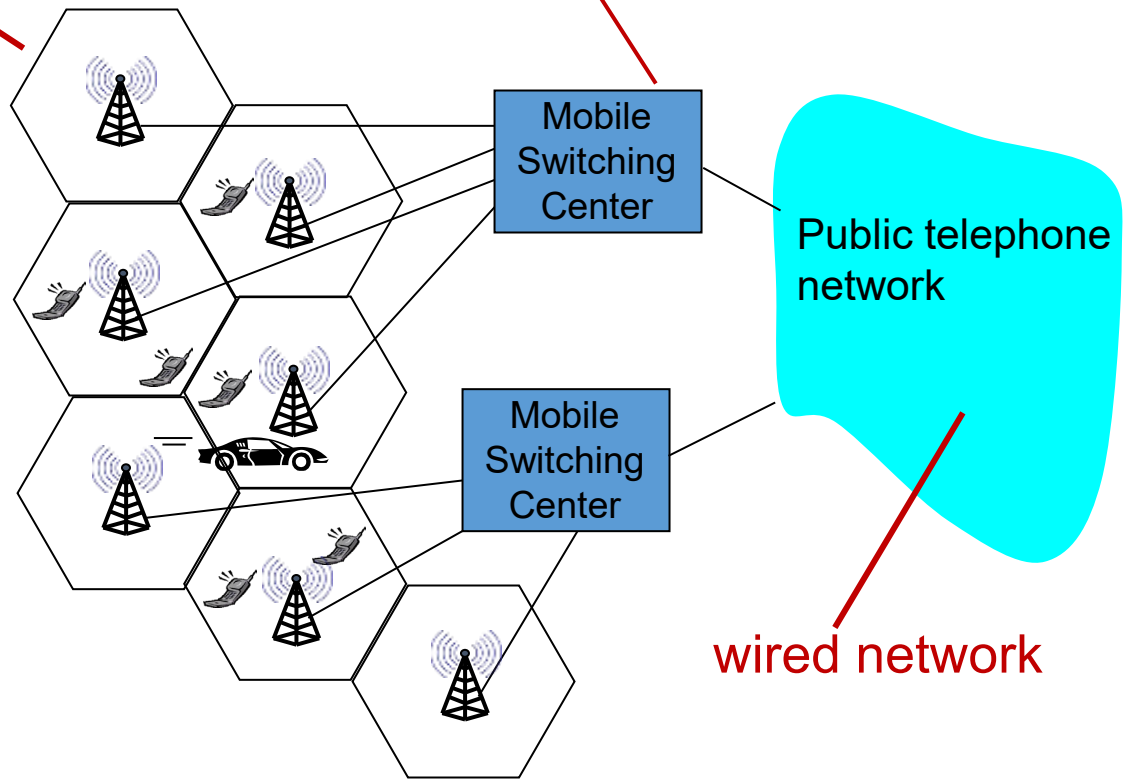
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Components of Cellular Network Architecture

- MSC: mobile switching center

MSC
- connects cells to wired tel. net.
- manages call setup (more later!)
- handles mobility (more later!)

cell
- covers geographical region
- *base station* (BS) analogous to 802.11 AP
- *mobile users* attach to network through BS
- *air-interface:* physical and link layer protocol between mobile and BS



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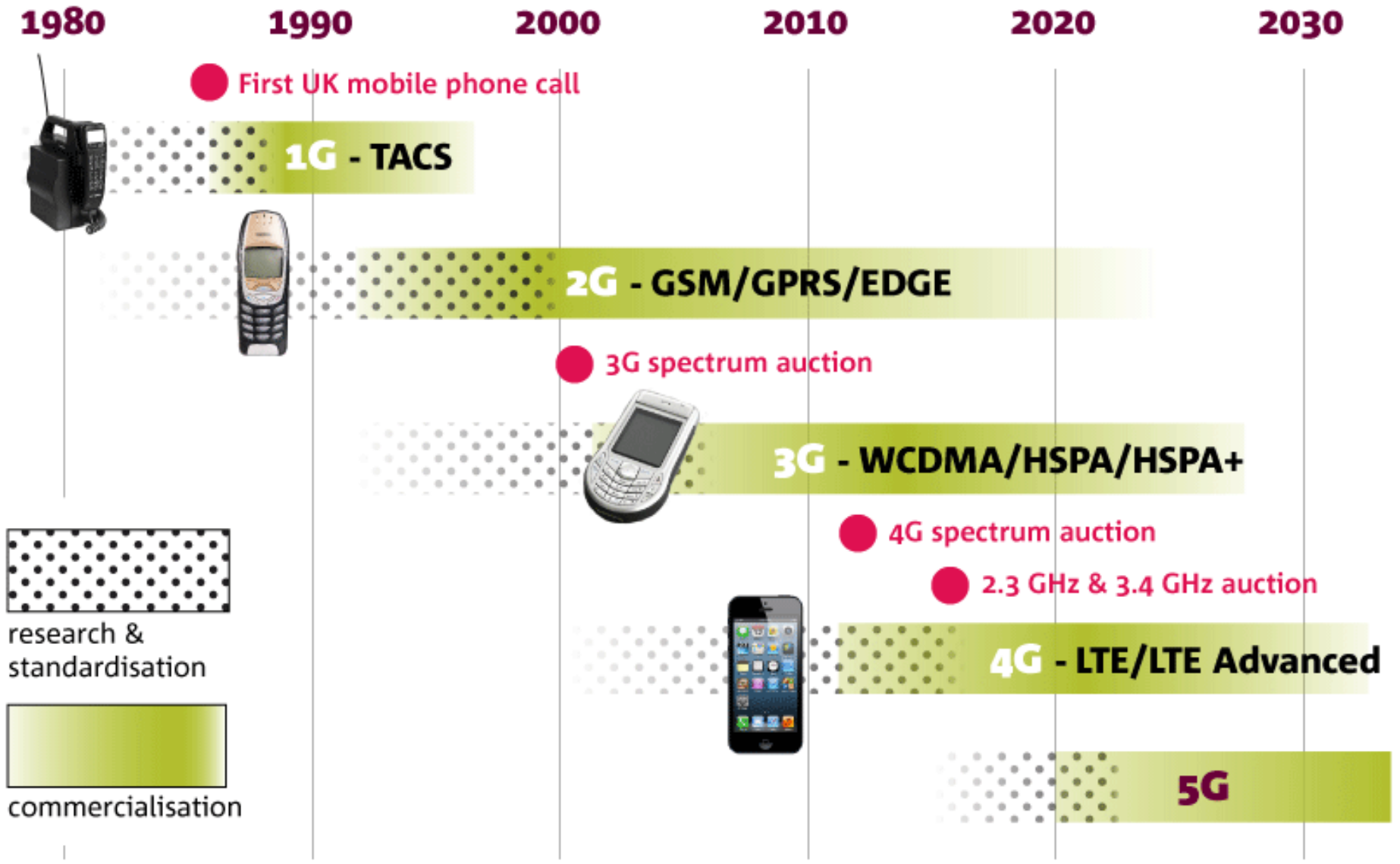
From 1G to 5G

: Their feature, Evolution, History of Korea Broadband

- Basics on mobile networks
- From 1G to 5G
- Fundamentals for mobile networks
- Summary & Next class

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[Review] Evolution of Mobile Communications



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Feature of 1G & 2G: Analog & Digital

1st commercially produced mobile phone in 1984

Retail price $9,000

Weighted 1Kg: brick phone



10 hours battery charge needed for only 1 hour talk time



AMPS, NMT, TACS

VS

Digital voice coding and modulation

D-AMPS, GSM/GPRS, CDMA



Deployed in early 1990s

100~200g hand-held devices

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Feature of 3G & 4G: Mobile Internet

MMS, Video Call and Video Streaming

Internet Service with Smartphone



WCDMA, CDMA2000, TD-SCDMA, Wibro

vs

LTE-Adv, Wibro-e



Anytime, anywhere

Mobile multimedia



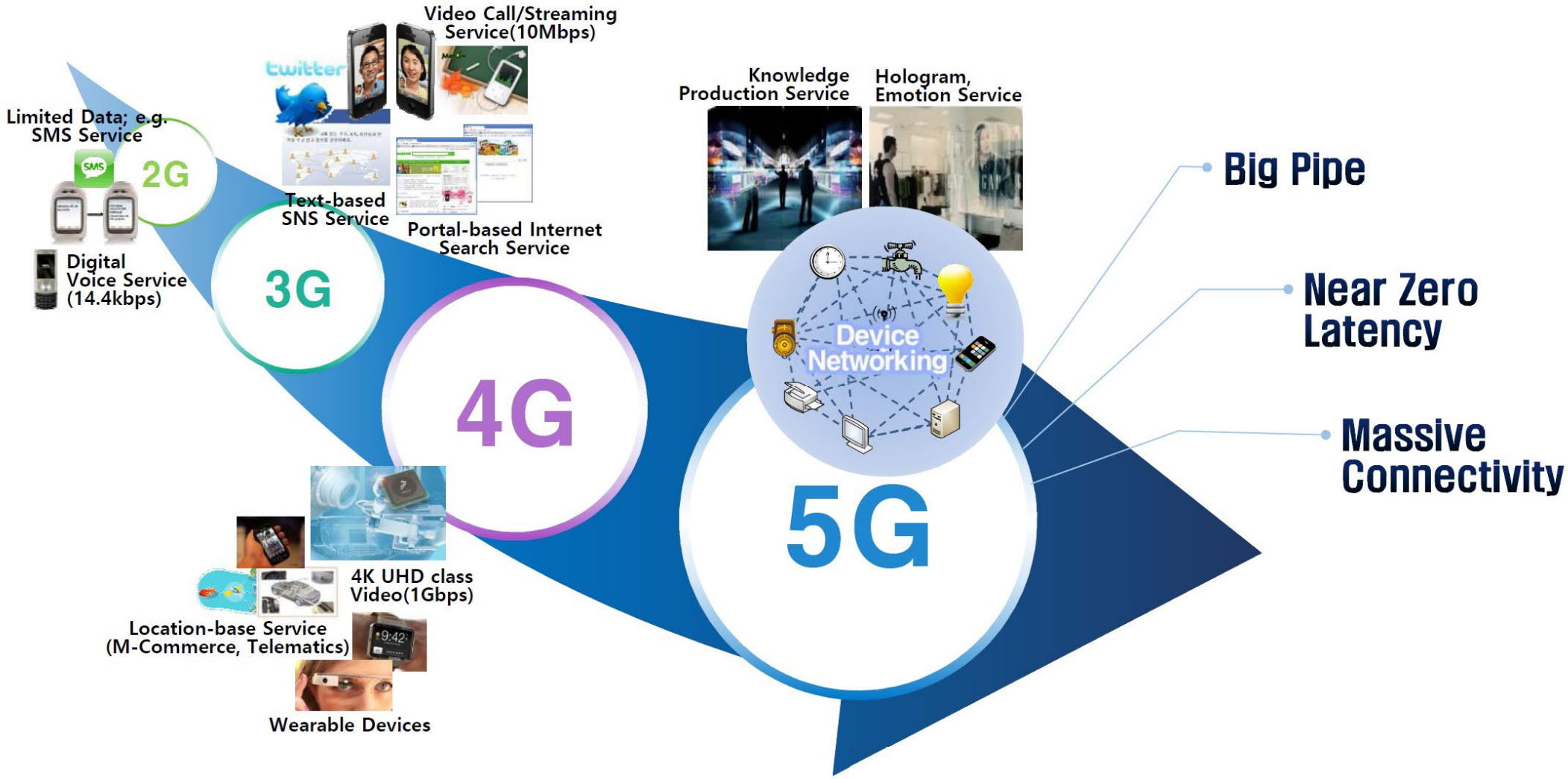
Integrated wireless solution

Customized personal services

Global mobility support

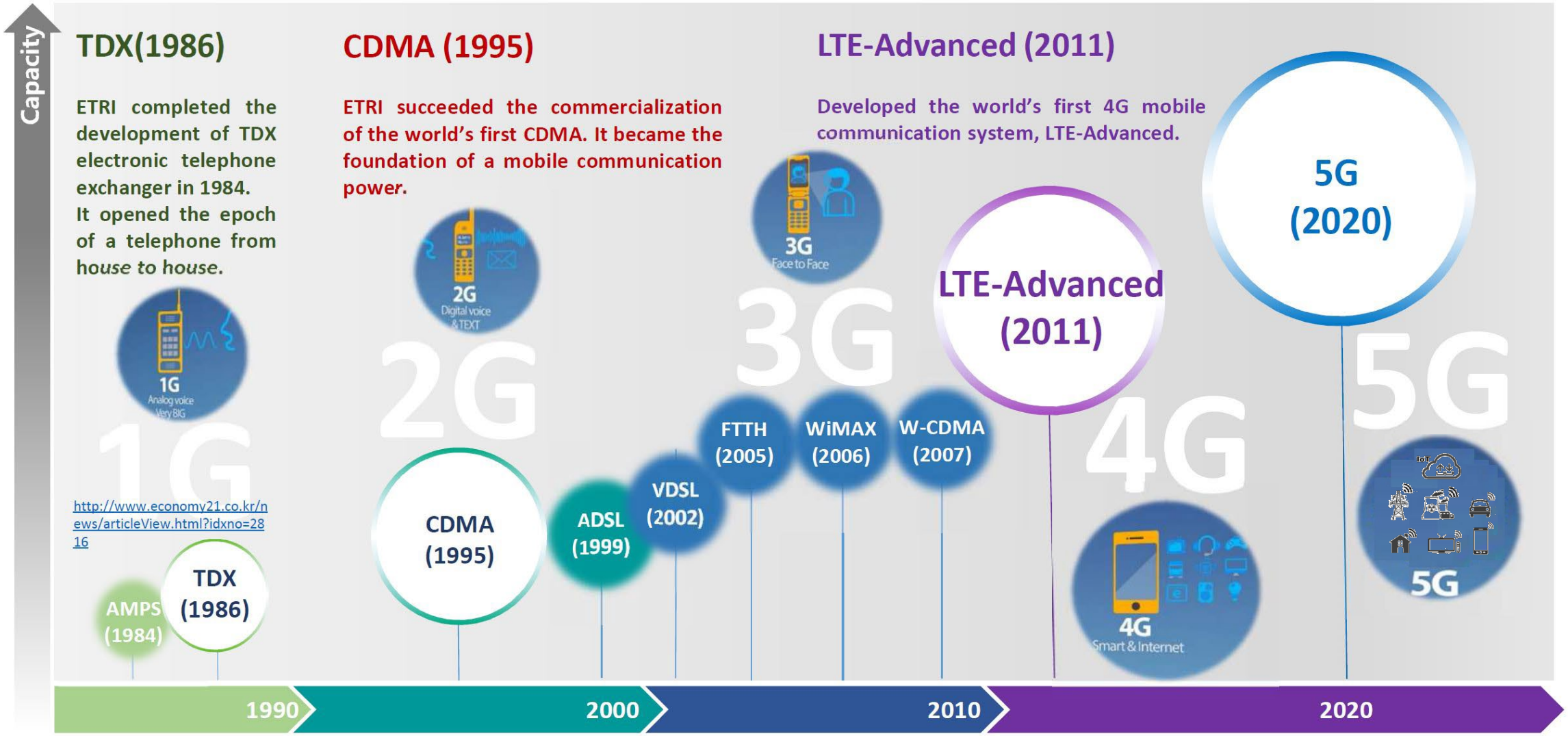
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Evolution of Mobile Communications: Services



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History of Korea's Broadband Services



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Fundamentals for Mobile Networks

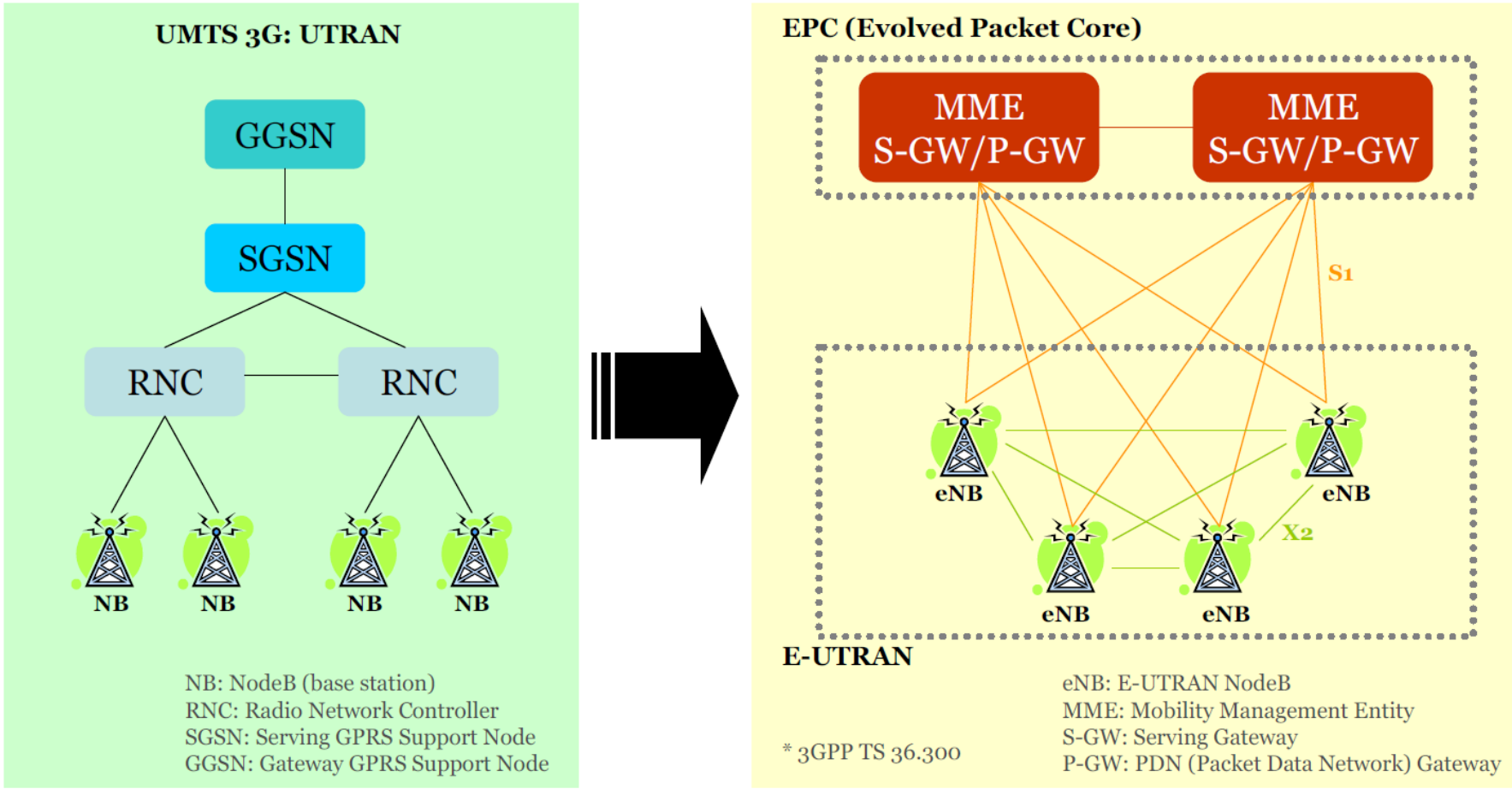
: Network architecture, Multiple access, Cell search/handover, 3GPP standards

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3G & 4G Network Architecture

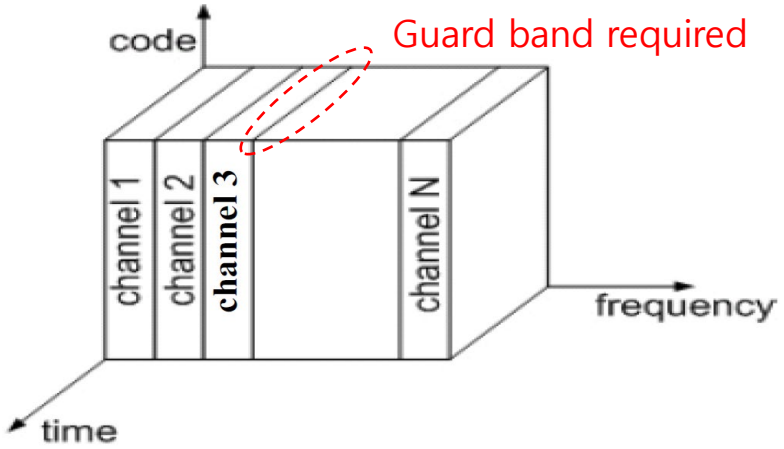
- E-UTRAN (Evolved Universal Terrestrial Radio Access Network)



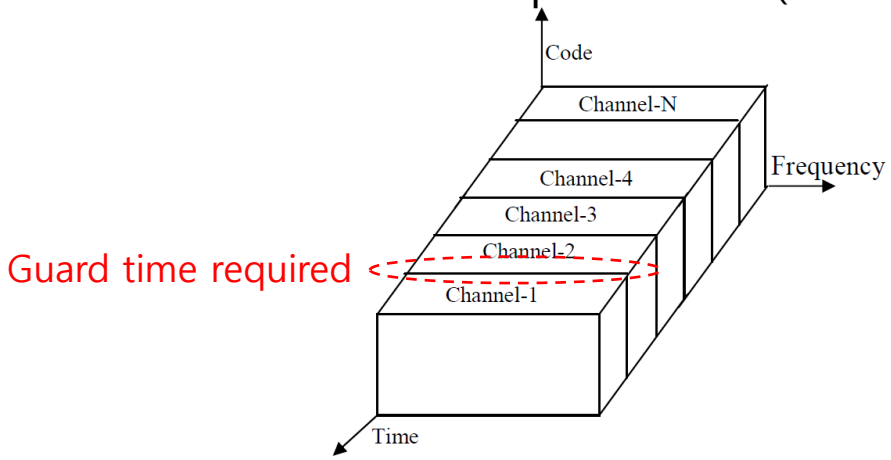
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Multiple Access for Multiuser Communications

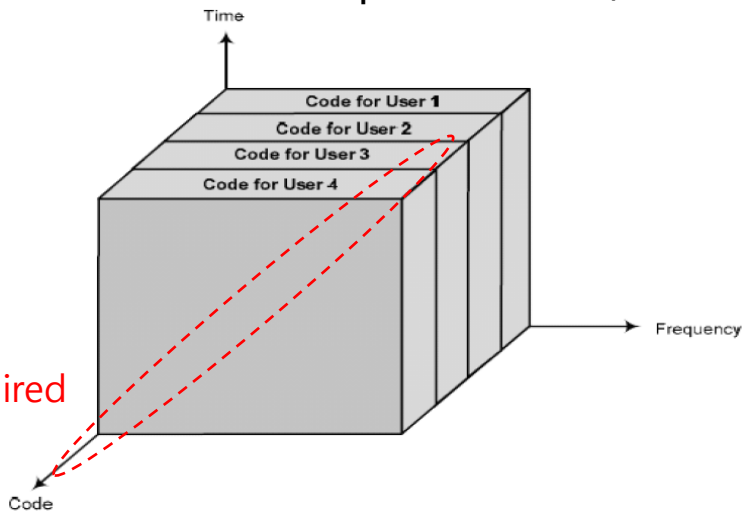
❑ Frequency Division Multiple Access (FDMA)



❑ Time Division Multiple Access (TDMA)



❑ Code Division Multiple Access (CDMA)



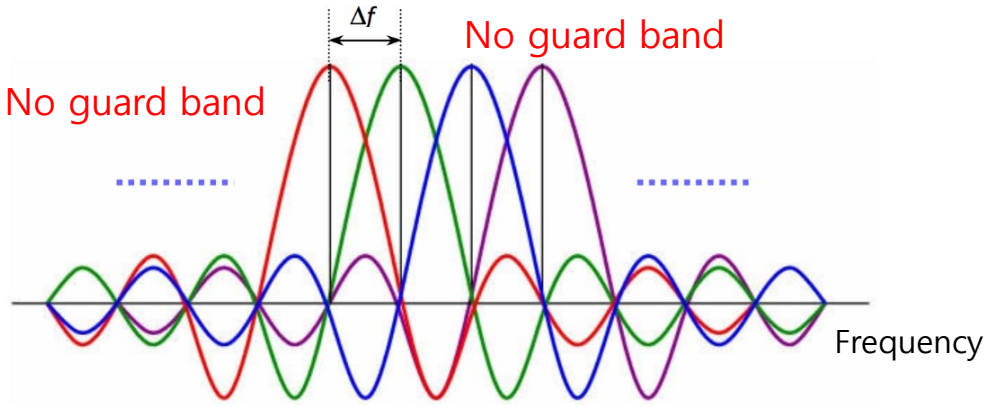
Much higher chip rate required
compared with data rate

FDMA/TDMA/CDMA :
Theoretically same capacity

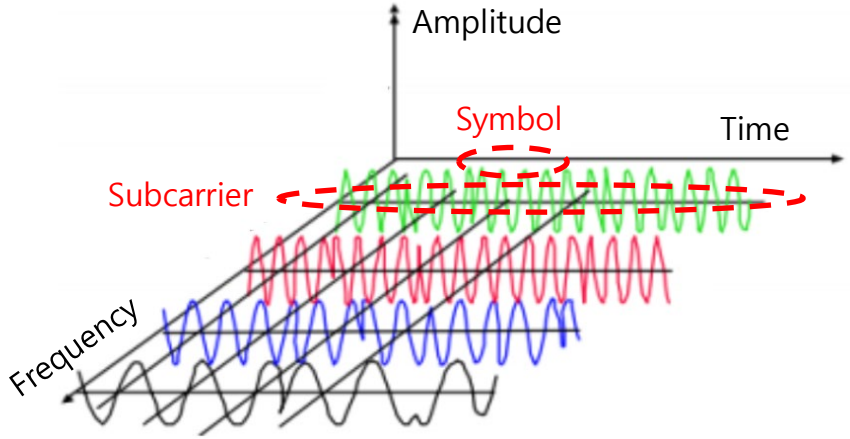
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OFDM for LTE and 5G NR

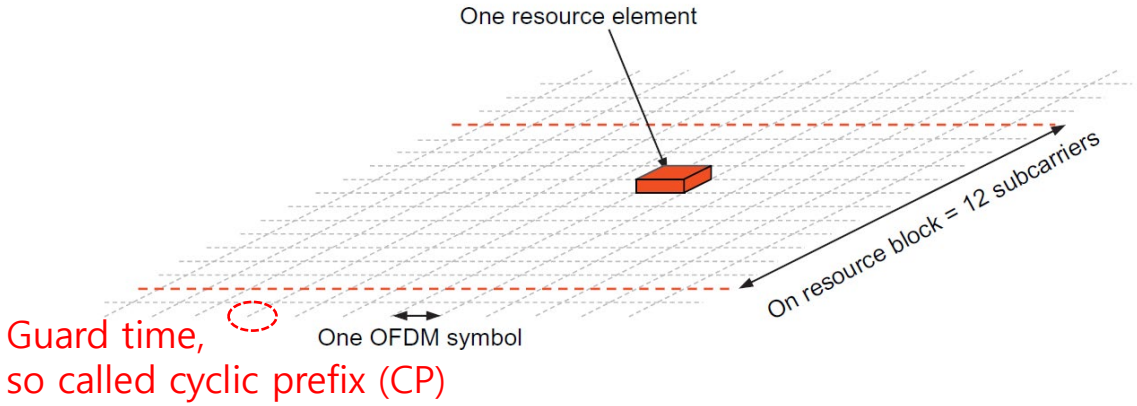
❑ Waveform in frequency domain



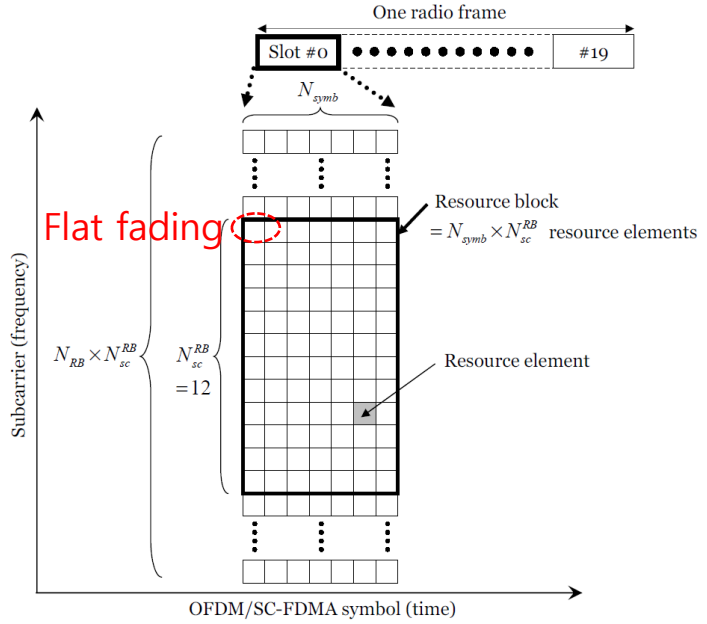
❑ Waveform in time domain



❑ Time-frequency resource for LTE and 5G



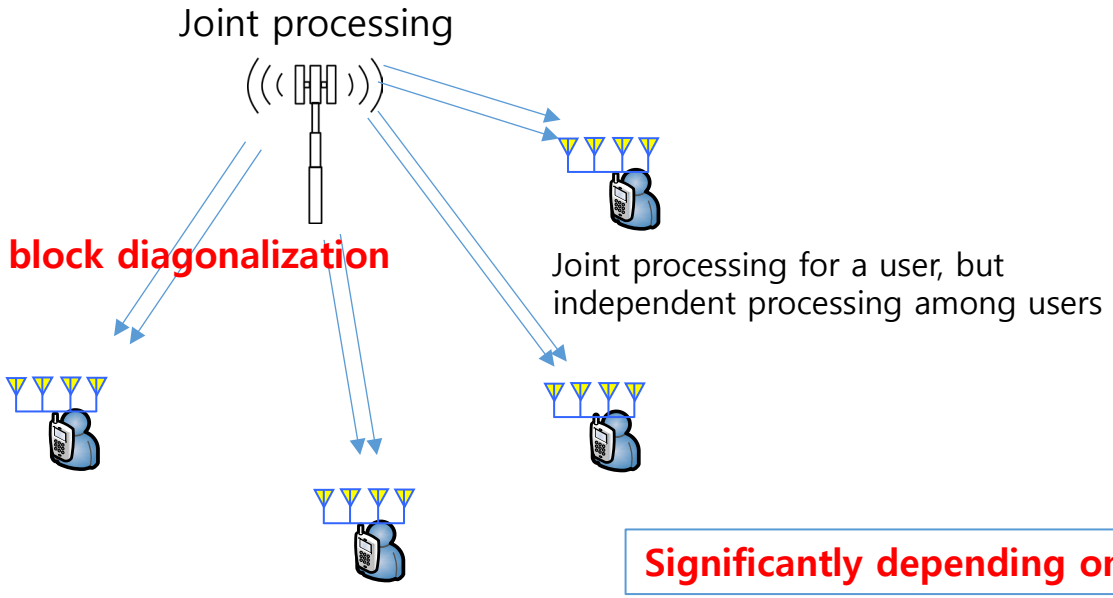
❑ E.g., LTE resource grid



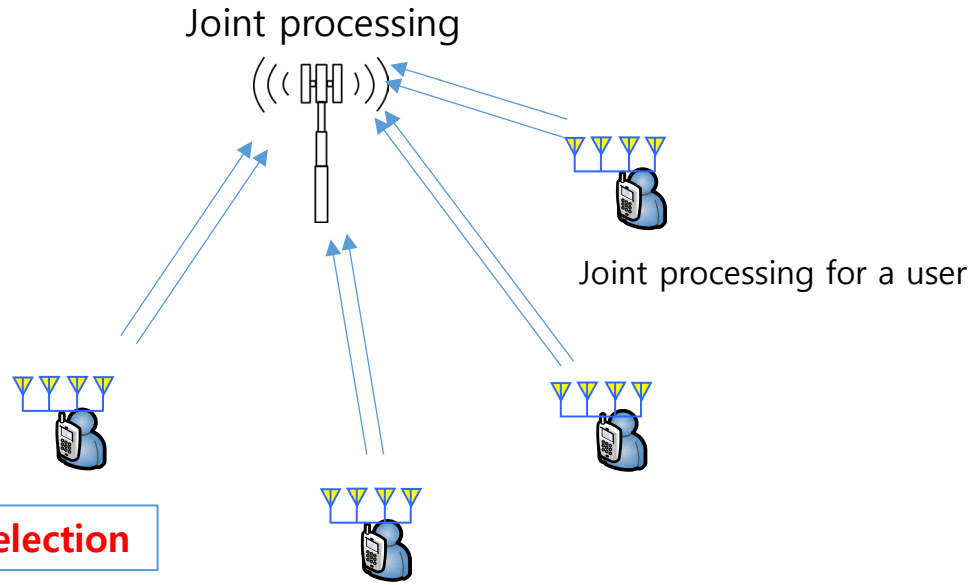
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Multiuser MIMO: MIMO + Scheduling

❑ Downlink



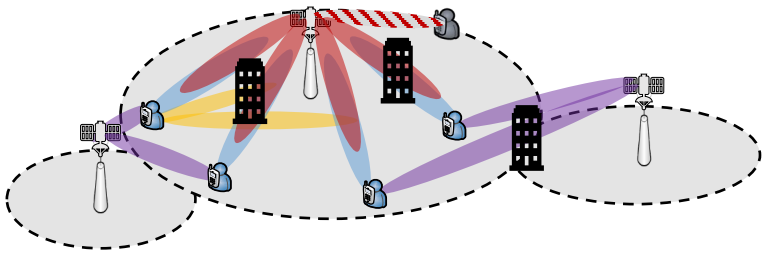
❑ Uplink



Significantly depending on user selection

❑ LTE-Advanced: downlink 8x8
- Typically, 4x2

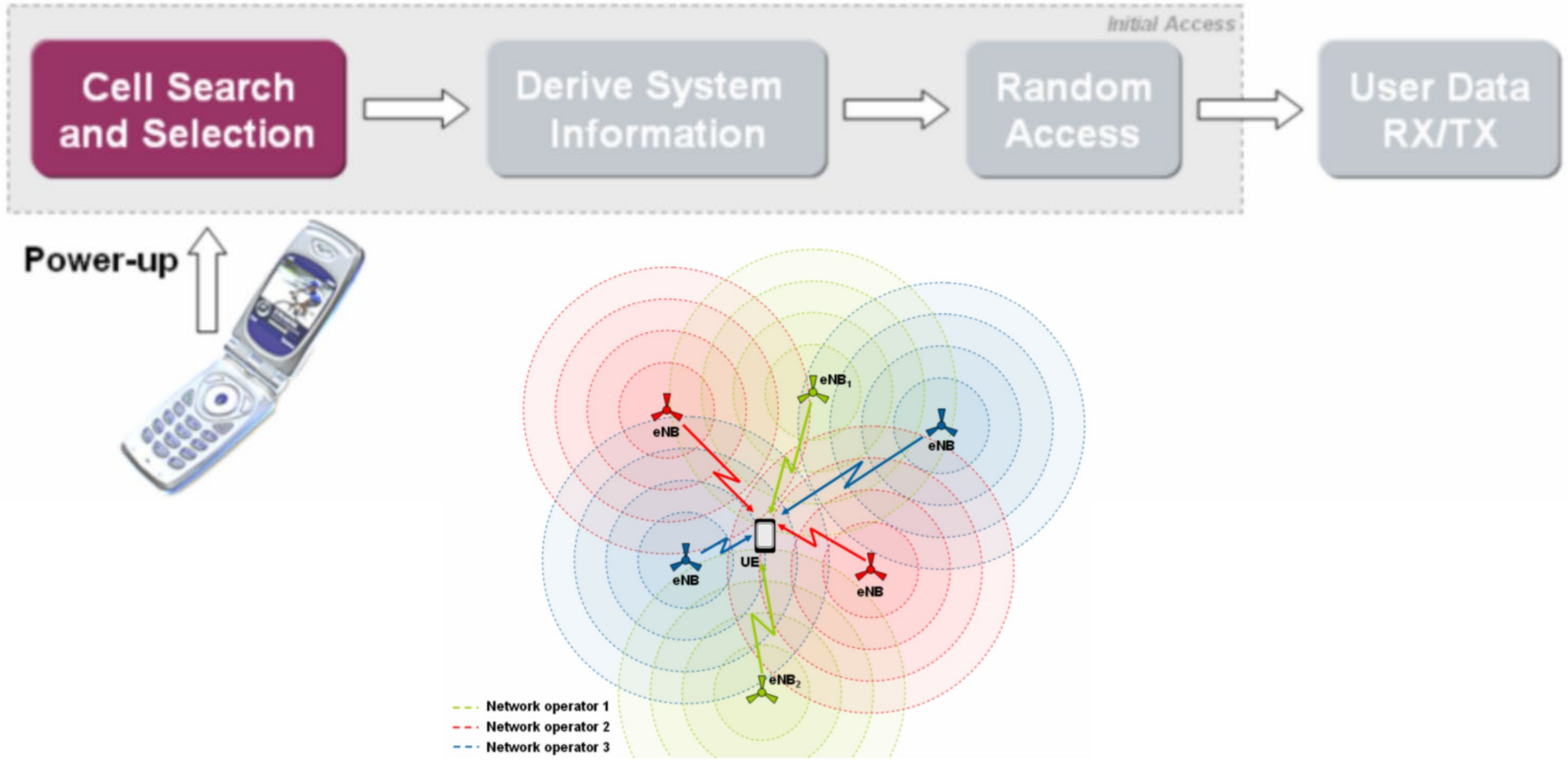
❑ 5G NR: 32 or 64 antennas at base station



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Cell Search

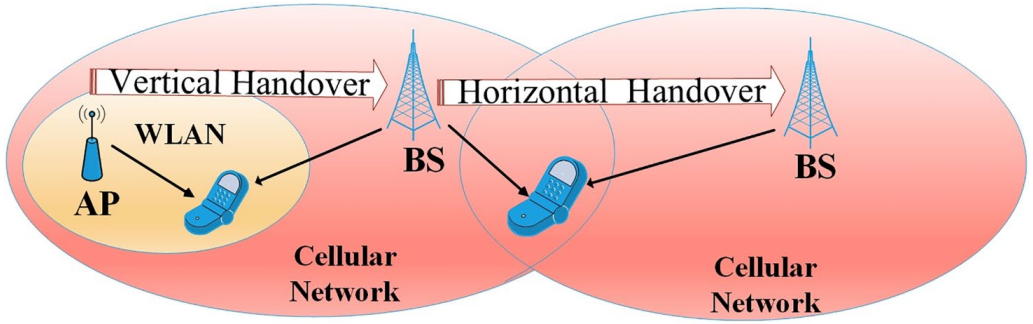
- General procedure for cell search and cell selection

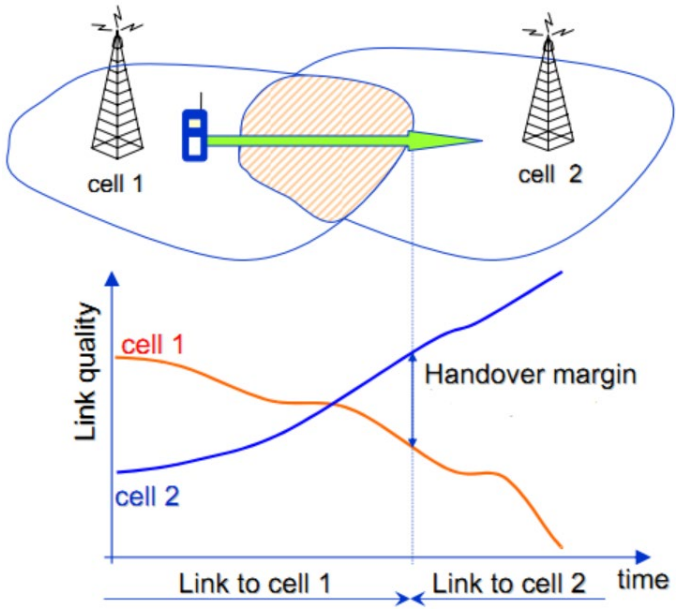


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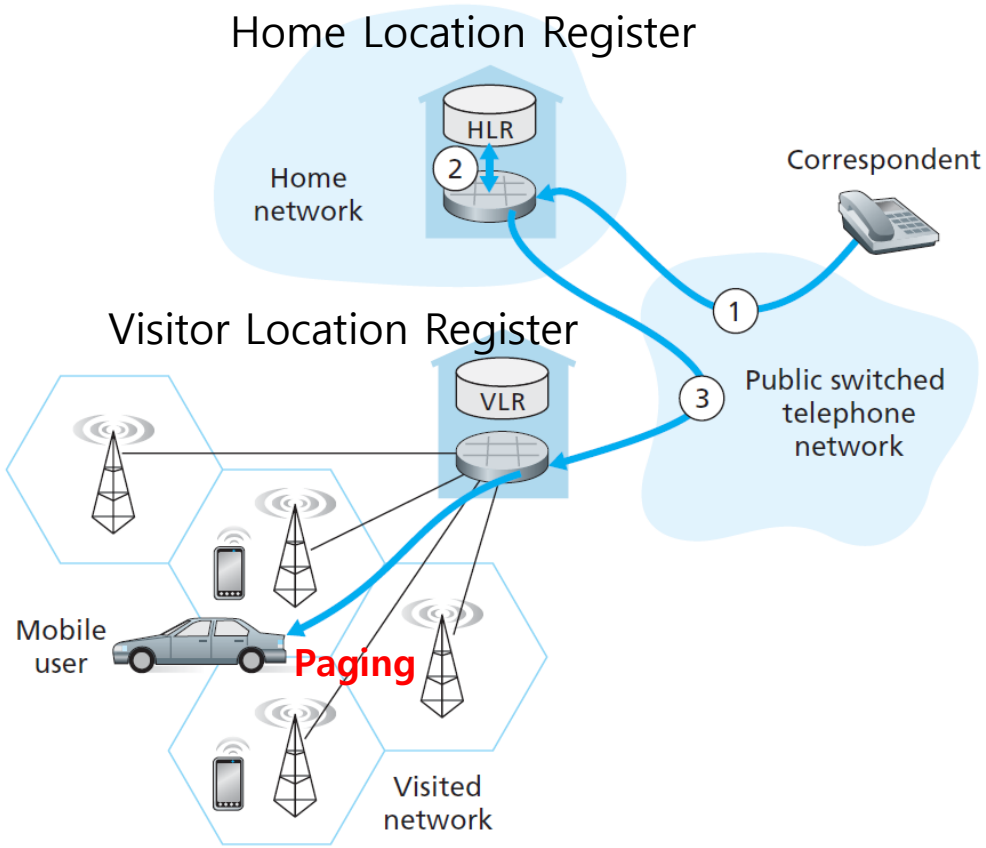
Handover

❑ Handovers





❑ General operations for supporting mobility



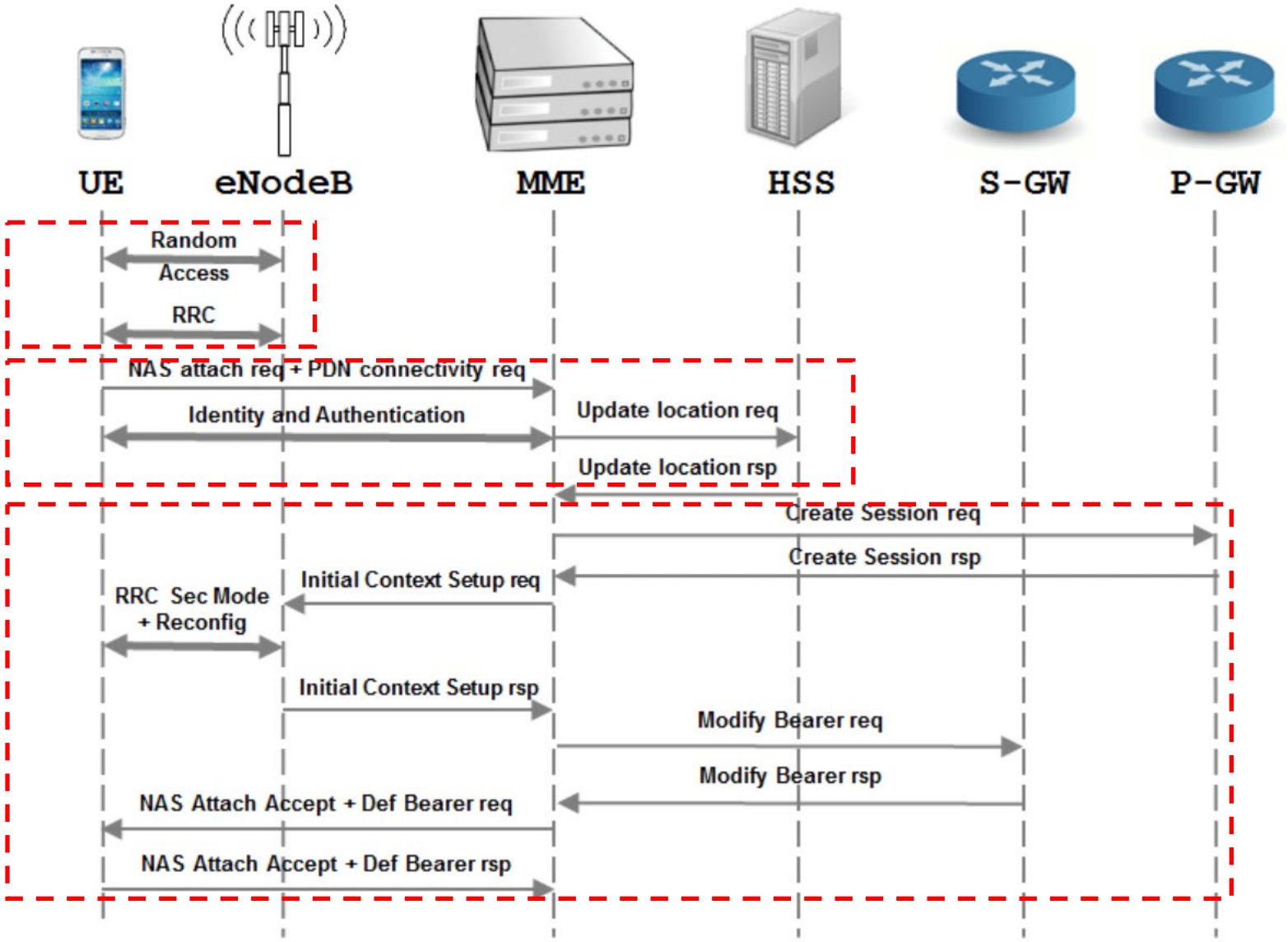
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Connection Setup to Access Network and Core Network

Connection to access network

Authentication and location update

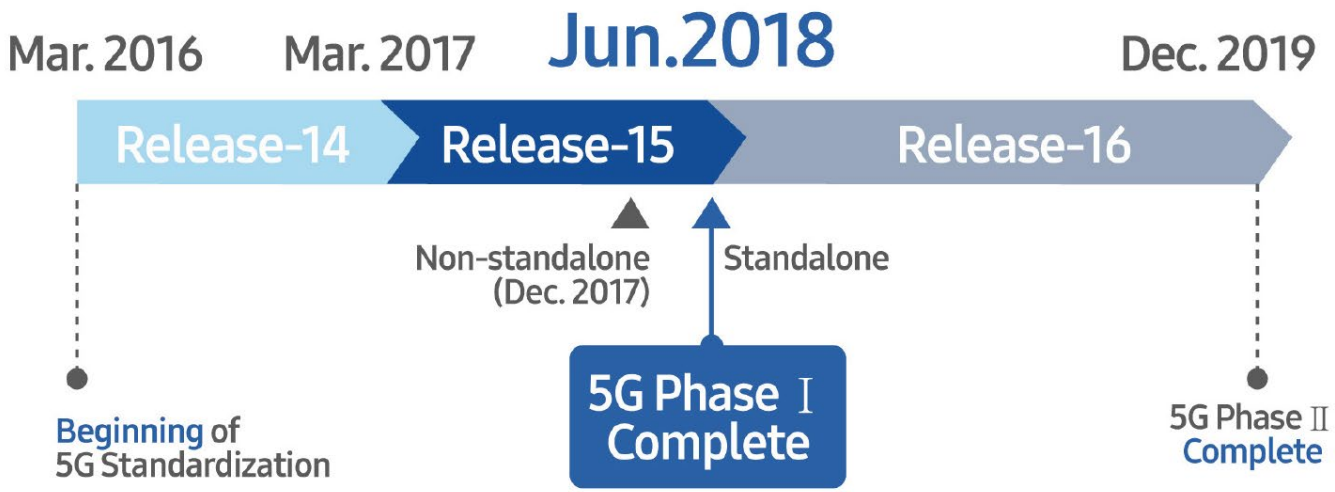
Connection to core network



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3GPP Standardization for 4G & 5G

- The 3GPP documents are divided into releases, where each release has a set of features added compared to the previous release
- The features are defined in Work Items agreed and undertaken by the TSGs.
- Key releases
  - Release 8: LTE
  - Release 10: LTE-Advanced (for IMT-Advanced)
  - Release 13: LTE-Adanced Pro (for 4.5G)
  - Release 15: 5G NR Phase 1
  - Release 16: 5G NR Phase 2

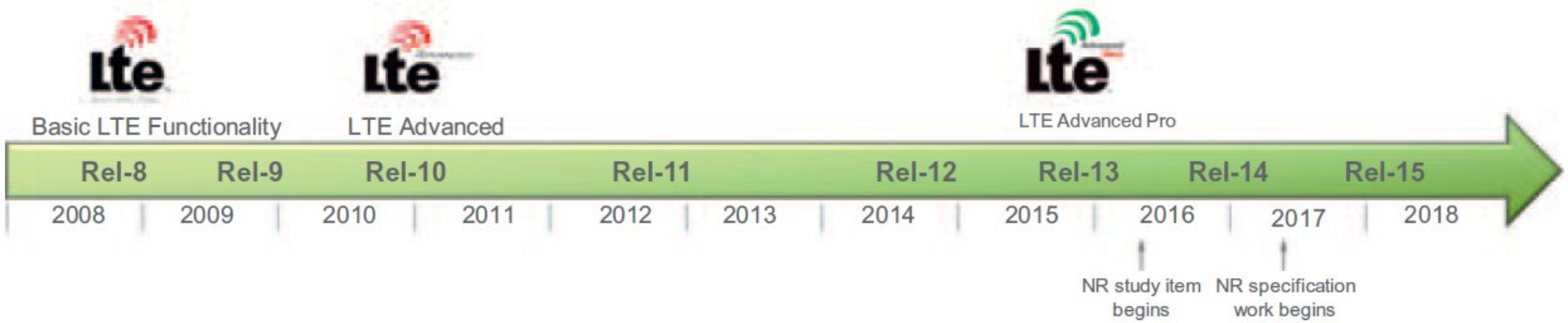


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3GPP Standardization for 4G & 5G

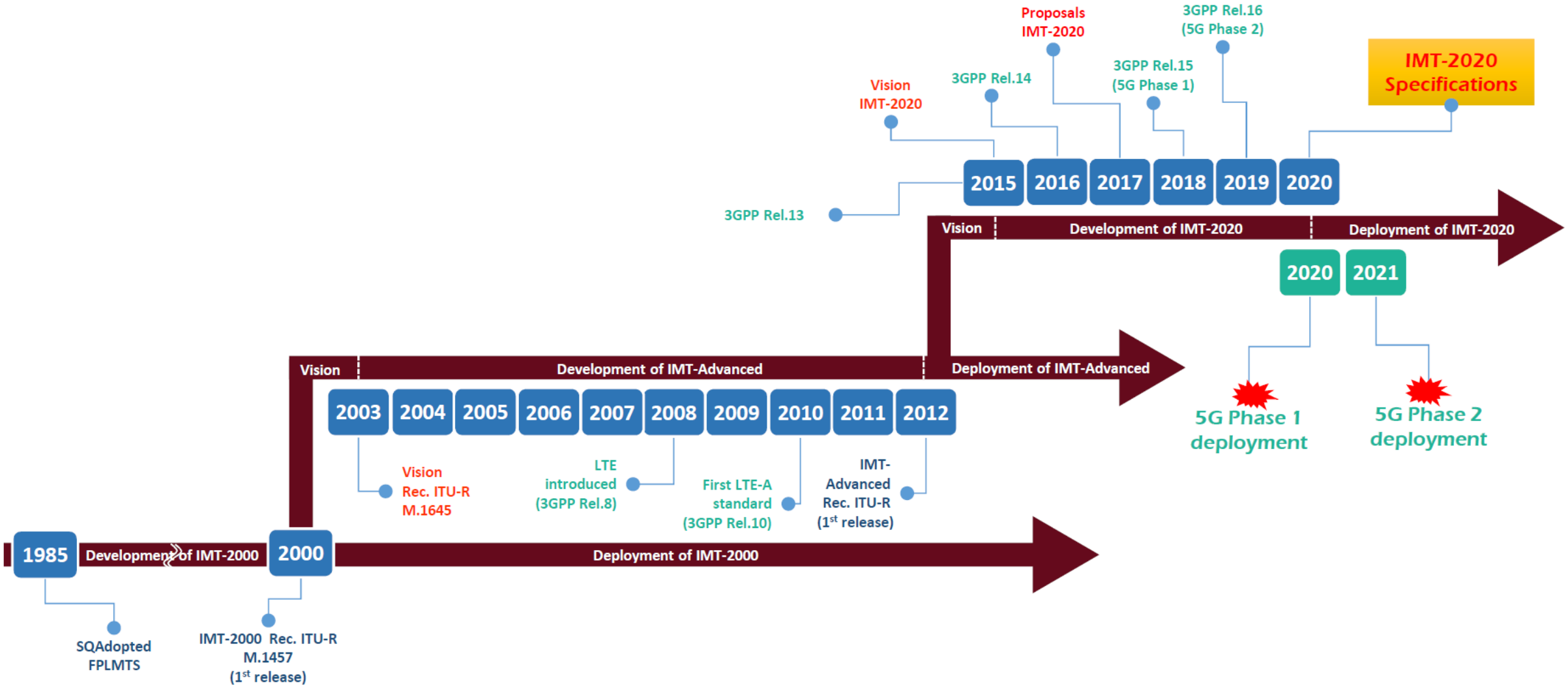
- LTE (Long Term Evolution) & NR
  - Developed by 3GPP
  - Common background and sharing several technology components, but not compatible
  - LTE continues to evolve in parallel with NR

- 3GPP standards for LTE
  - Initiated in late 2004 for a new radio-access technology focusing on packet-switched data only
  - First release of LTE spec: Release 8 completed in 2008, commercialized in late 2009
  - LTE → LTE-Adv (4G) → LTE-Adv Pro (4.5G)



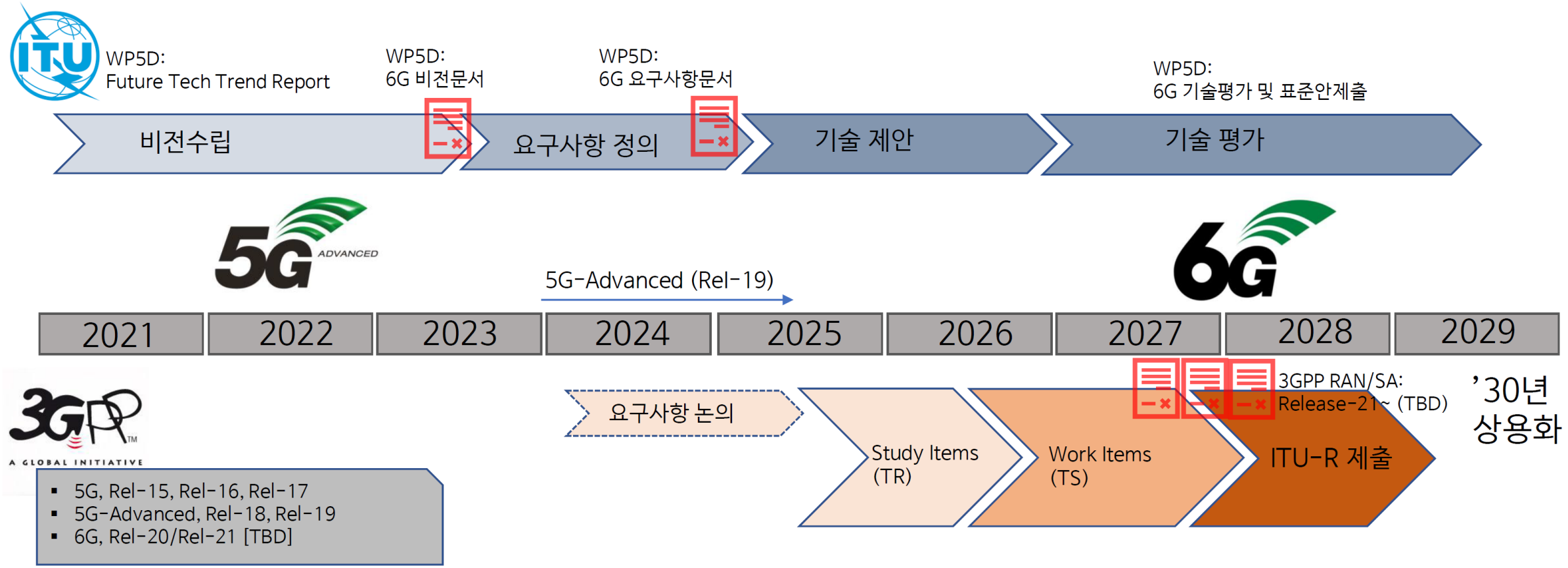
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3GPP Standard Evolution



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[Review] 5G-Adv & 6G Timeline [TBD]



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Series of 3GPP Specifications

- Available on https://www.3gpp.org

- Example

Series	Description
25-series	• Radio aspects for UTRA (WCDMA/HSPA)
45-series	• Radio aspects for GSM/EDGE
36-series	• Radio aspects for LTE, LTE-Advanced and LTE-Advanced Pro
37-series	• Aspects relating to multiple radio access technologies
38-series	• Radio aspects for NR

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Summary & Next Class

- Basics on mobile networks
- From 1G to 5G
- Fundamentals for mobile networks
- Summary & Next class

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Summary

- Basics on mobile networks
  - Wireless network elements: BS, UE, link
  - Infrastructure mode vs. ad hoc mode, single hop vs. multi-hop, adapting MCS, handover
- From 1G to 5G
  - 1G: analog voice, 2G: digital voice & SMS, 3G: MMS/video call/Internet
  - 4G: high data rate services, 5G: eMBB/URLLC/mMTC
- Fundamentals for mobile networks
  - 4G terminologies: E-UTRAN, EPC, eNB, UE
  - OFDM, MIMO, cell search, handover, 4G/5G/6G 3GPP standards

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Course Schedule (Tentative)

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15	**Final exam**	**Class hours**	**12/10**

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